Name: ___________________________

Date: _____________________________

Mr. Rawson • Y8 MYP Physics

1.11 – Fluorescence & Phosphorescence

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain the difference between fluorescence and phosphorescence based on the time delay of light emission.
- b) I can describe how materials absorb high-energy ultraviolet (UV) light and re-emit it as visible light.
- c) I can identify real-world applications of these phenomena, such as security features and forensic science.

Theory

When certain materials are exposed to **ultraviolet (UV)** light, which is a type of electromagnetic radiation with higher energy than visible violet light, they absorb that energy. In the case of **fluorescence**, the material absorbs the UV photons and almost instantly re-emits the energy as visible light. This process happens so quickly that the glow stops the moment the UV source is removed. The emitted light always has less energy (and a longer wavelength) than the absorbed light, which is why the colour changes to something we can see, often appearing bright and vibrant under a blacklight.

Phosphorescence works on a similar principle of absorbing energy but involves a crucial difference: time. In phosphorescent materials, electrons get trapped in a temporary energy state after absorbing photons. They release this stored energy very slowly over time, causing the material to continue glowing even after the light source has been turned off. This is the mechanism behind **glow-in-the-dark** stickers and watches that you charge under a lamp before sleeping. While fluorescence is immediate, phosphorescence is a delayed reaction.

These properties are not just party tricks; they have vital scientific and commercial uses. For example, many banknotes contain **security markings** made of fluorescent inks that are invisible under normal light but glow brightly under UV lamps to prove authenticity. In medicine and crime scene investigation, **forensics** teams use fluorescent dyes to detect blood traces or fingerprints that the naked eye cannot see. Additionally, washing powders often contain **optical brighteners** which absorb UV light from sunlight and re-emit it as blue visible light, making white clothes appear brighter and whiter.

Word bank: phosphorescence · fluorescence · ultraviolet · security markings · optical brighteners

Activity

The Blacklight Investigation

You will need a UV torch (or blacklight), a sheet of white paper, a highlighter pen (fluorescent yellow or green), and a glow-in-the-dark sticker (phosphorescent).

- Turn off the room lights to make the environment as dark as possible.
- Shine the UV torch onto the white paper. Observe if any light is emitted.
- Draw a line on the paper using the fluorescent highlighter. Shine the UV torch on the drawing. Describe the colour and intensity of the glow compared to the pen in normal daylight.
- Place the glow-in-the-dark sticker under the UV torch for 10 seconds, then remove the torch immediately. Observe the sticker while it is still being lit, then turn the torch away completely.

- Record your observations in the table below.

Material	Immediate Glow (while light is on)?	Delayed Glow (after light is off)?	Ty
White Paper	No	No	
Highlighter Line	Yes	No	
Glow Sticker	Yes	Yes	

Questions

1. Explain why the fluorescent highlighter appears brighter under UV light than in normal room lighting, referring to energy absorption and re-emission.

2. A scientist uses a UV lamp to examine a white shirt for blood stains. The blood glows blue-green under the lamp but is invisible in daylight. Is this an example of fluorescence or phosphorescence? Give one reason for your answer.

3. Why can we not see the security markings on a banknote using our naked eyes in a normal shop, but they are clearly visible with a UV torch?

4. If you leave a fluorescent pen cap in direct sunlight for an hour and then take it into a dark room, will it continue to glow after the sun is gone? Explain your reasoning based on the difference between fluorescence and phosphorescence.

Extension

Find out what **optical brighteners** are commonly added to laundry detergents. Research why they might be harmful to aquatic life if they enter rivers and streams in large quantities, and suggest one alternative way to make whites look brighter without using these chemicals.